PYTHON FILE AUTOMATION SCRIPT

March 24, 2026 (Piyush Patil)

OVERVIEW

1. Project background and description

Managing files manually can be time-consuming and inefficient. This project focuses on automating file organization using Python. The script scans a user-specified folder, identifies file types, and moves them into appropriate categories such as Images, Documents, Videos, and Audio. It also maintains a log file to track all operations performed.

2. Project scope

This project can be extended to support more file types and advanced features like automatic scheduling, duplicate file detection, and integration with cloud storage systems for better file management.

3. Objective

To develop a Python script that automates the process of organizing files into different folders based on their types.

4. Features

- *Automatic file organization*
- *Categorization based on file types (Images, Documents, Videos, Audio)*
- *User input for folder path*
- *Logging of file movements and skipped files*
- *Error handling using exception handling*

5. Technologies Used

- Python
- OS module
- Shutil module
- Datetime module

6. Steps Performed

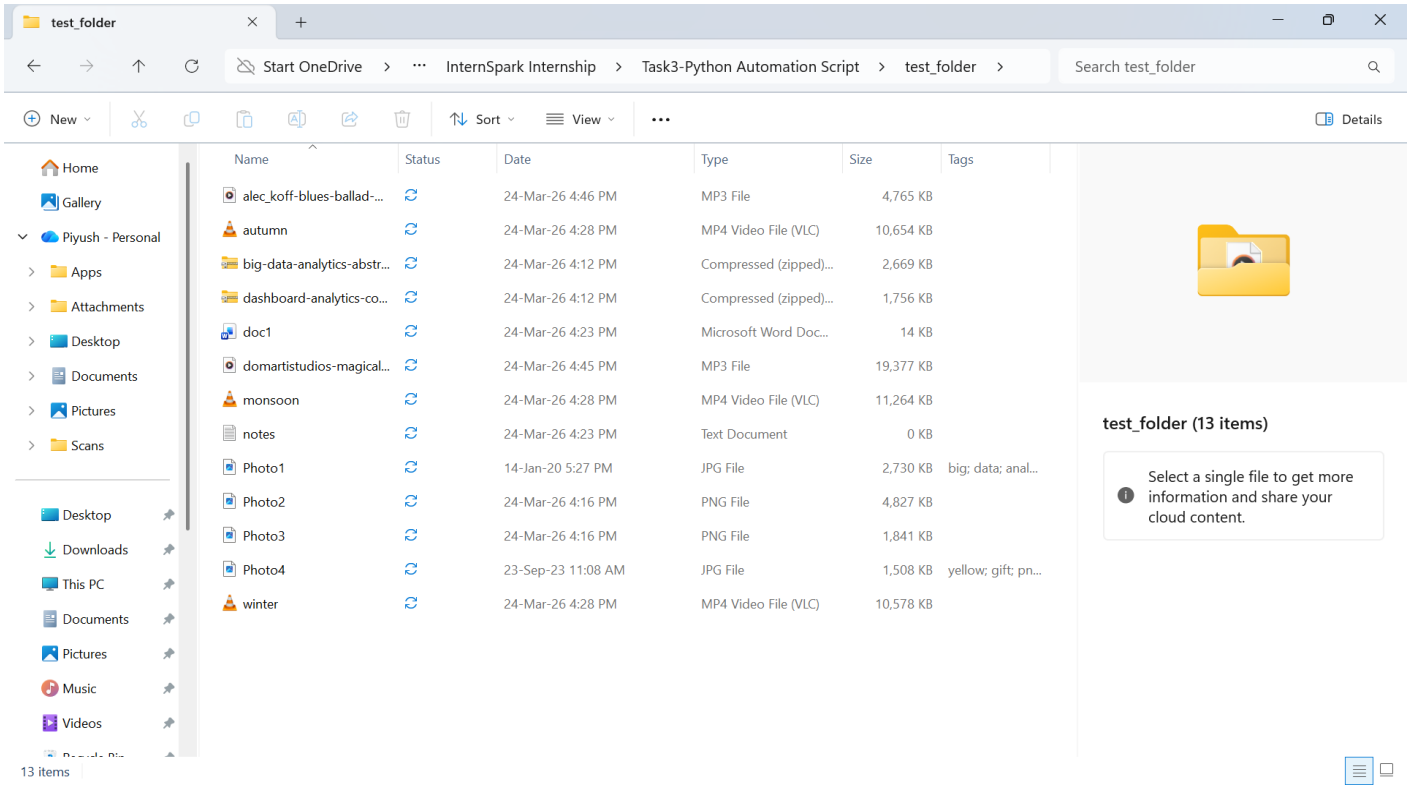
- Accepted folder path as input from the user
- Scanned all files in the folder
- Identified file extensions
- Created folders dynamically based on file types
- Moved files into respective folders
- Logged all actions in a log file
- Handled errors using exception handling

7. Deliverables

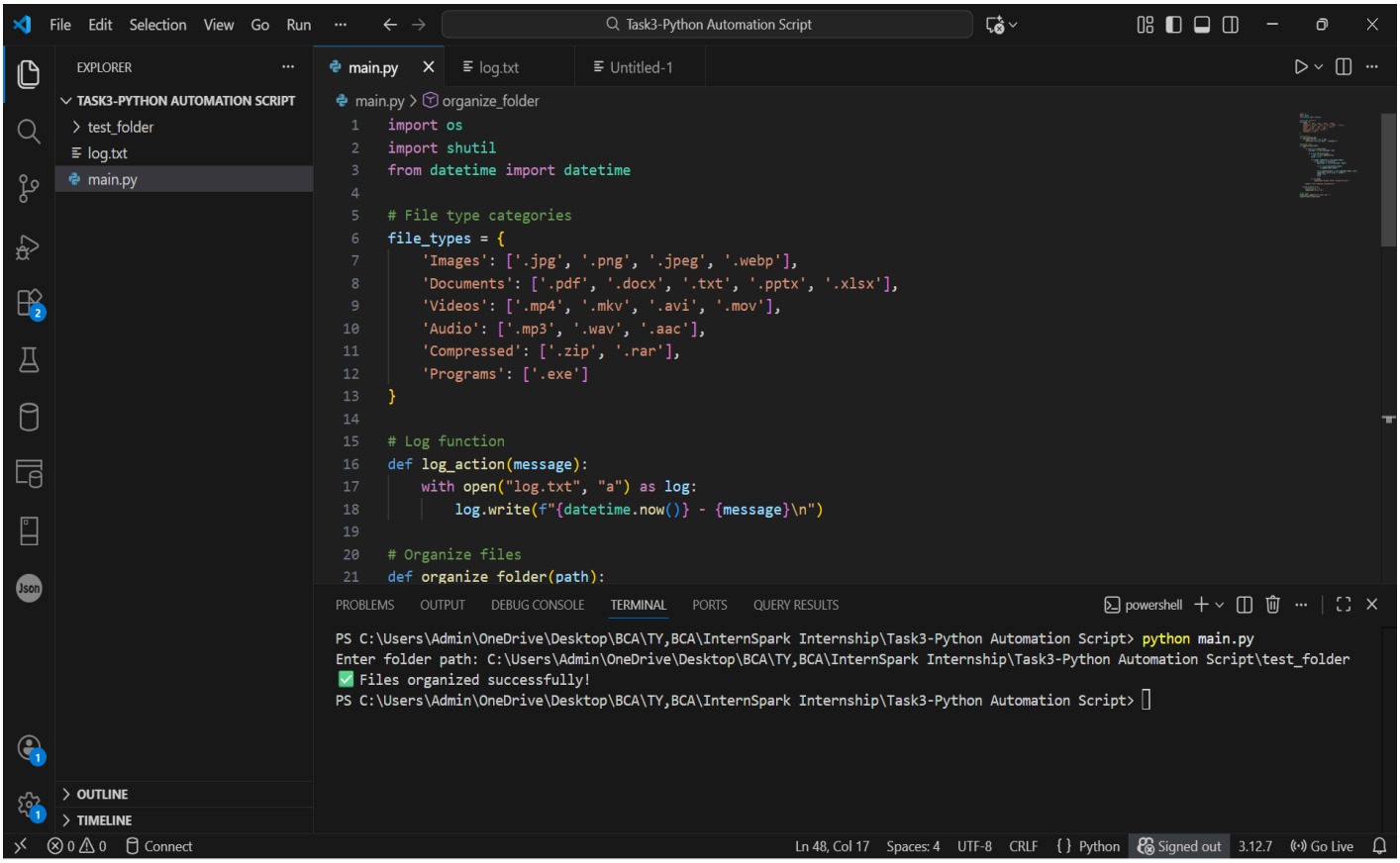
- Python automation script
- Sample input/output screenshots
- Log file
- Project documentation

SCREENSHOTS:

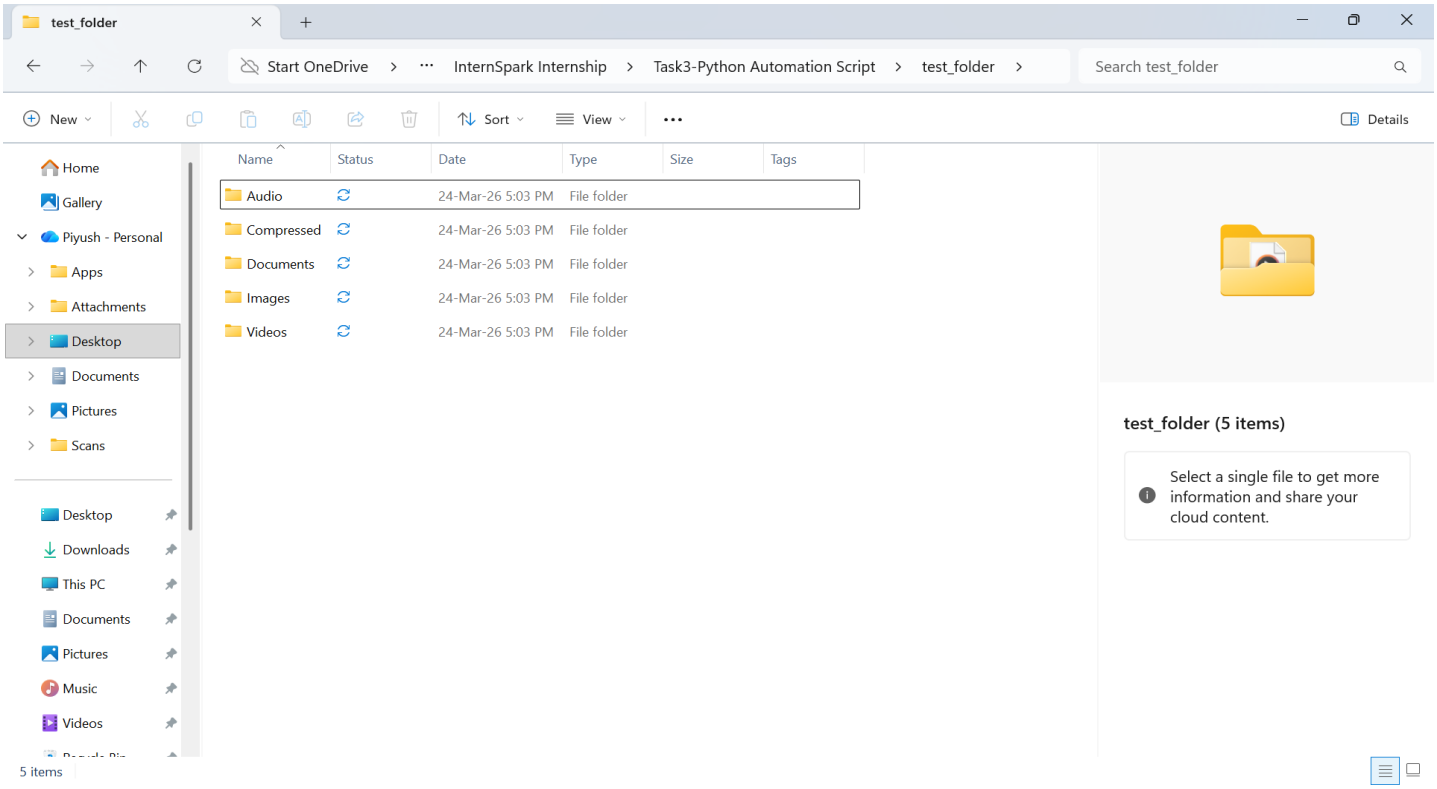
BEFORE (messy folder)



PROCESS (terminal running)



AFTER (organized folder)



RESULT (log file)

log.txt

File Edit View

```
2026-03-24 17:03:16.114855 - Moved alec_koff-blues-ballad-487408.mp3 to Audio
2026-03-24 17:03:16.116855 - Moved autumn.mp4 to Videos
2026-03-24 17:03:16.119848 - Moved big-data-analytics-abstract-concept-illustration.zip to Compressed
2026-03-24 17:03:16.121847 - Moved dashboard-analytics-computer-performance-evaluation-chart-screen-statistics-analysis-infographic-ass.zip to Compressed
2026-03-24 17:03:16.123353 - Moved doc1.docx to Documents
2026-03-24 17:03:16.124361 - Moved domartistudios-magical-wizard-school-orchestral-fantasy-488126.mp3 to Audio
2026-03-24 17:03:16.125366 - Moved monsoon.mp4 to Videos
2026-03-24 17:03:16.126364 - Moved notes.txt to Documents
2026-03-24 17:03:16.128364 - Moved Photo1.jpg to Images
2026-03-24 17:03:16.129365 - Moved Photo2.png to Images
2026-03-24 17:03:16.130365 - Moved Photo3.png to Images
2026-03-24 17:03:16.131364 - Moved Photo4.jpg to Images
2026-03-24 17:03:16.132364 - Moved winter.mp4 to Videos
```

8. Conclusion

This project demonstrates how Python can be used to automate repetitive tasks like file organization. It improves efficiency, reduces manual effort, and provides a structured way to manage files effectively.

9. How to Run the Project

1. Run python app.py (in terminal)
2. Enter the path of folder in it (the folder which needs to be sorted)